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MANIPAL UNIVERSITY
JAIPUR

Odd Semester Mid Term Examination, Academic Session: Jul-Nov 2026
FoSTA, School of Computer Science and Engineering
Department of AIML
B. Tech CSE (AI&ML)

Course Code: AIM3102

Course: Automata Theory & Compiler Design

Semester: V

Time: 1.5 hrs.

Max. Marks: 30

Instructions: All questions are compulsory.
Missing data, if any, may be assumed suitably.
Calculator is not allowed.

SECTION A

S.No.		Marks	CO
Q 1	Explain and find regular expression for the following DFA using Arden's Theorem and explain Arden's theorem in detail.	2	CO1
Q 2	Explain PDA with example and its type. Also Consider the following language: $L_1 = \{a^n b^m c^n : m, n \geq 1\}$ write down the type of language and grammar for given language.	2	CO2
Q 3	Construct a right-linear grammar $G = (V, T, P, S)$ for $L = \{w \in \{a, b\}^* \mid w \text{ contains the substring } ab\}$ and show a leftmost derivation of the string "aabb".	2	CO3
Q 4	Illustrate and design a DFA with $\Sigma = \{0, 1\}$ for the language accepting strings starting and ending with different characters. Also write transition function (δ) and Table for given DFA.	2	CO1

SECTION B

Q 5	(a) Compare DFA, NFA and convert the following Non-Deterministic Finite Automata (NFA) to Deterministic Finite Automata (DFA). Explain step by step and write transition table and transition function for DFA. (b) Solve which one of the following languages over $\Sigma = \{a, b\}$ is NOT context-free? (A) $\{ww^R \mid w \in \{a, b\}^*\}$ (B) $\{wa^n b^n w^R \mid w \in \{a, b\}^*, n \geq 0\}$ (C) $\{wa^n w^R b^n \mid w \in \{a, b\}^*, n \geq 0\}$ (D) $\{a^n b^i \mid i \in \{n, 3n, 5n\}, n \geq 0\}$	5 (4+1)	CO2
Q 6	(a) Illustrate the working principle of a Turing Machine. Describe the role of the tape, tape head, transition function, and states. And explain unrestricted grammar. (b) Correlate the Chomsky Hierarchy with a neat diagram. Discuss the four types of grammar and their corresponding computational models. Differentiate between Type-0, Type-1, Type-2, and Type-3 grammar with suitable examples.	5 (2.5+2.5)	CO3

Q 7	(a) Evaluate and consider the given grammar- $S \rightarrow A1B$ $A \rightarrow 0A / \epsilon$ $B \rightarrow 0B / 1B / \epsilon$ For the string $w = \underline{00101}$, find- Leftmost derivation, Rightmost derivation and Parse Tree (b) Convert the given CFG to CNF. Consider the given grammar G: $S \rightarrow a \mid aA \mid B$ $A \rightarrow aBB \mid \epsilon$ $B \rightarrow Aa \mid b$	5 (2.5+2.5)	CO2
SECTION-C			
Q 8	(a) Construct a PDA for $L = \{wcw^R \mid w \in (a+b)^*\}$, where w^R is a reverse of w. Give its formal definition, the transition rules and the sequence of instantaneous descriptions (IDs) for the input string $w=bbaabb$. Further to check, is w is accepted by PDA? (b) Design a Turing machine that decides $L = \{a^n b^n \mid n \geq 1\}$. Show that an ID for the string "aaabbb" with the tape symbols and transition table.	4 (2+2)	CO2, CO3
Q 9	(a) Create a DFA over $\Sigma = \{0, 1\}$ that accepts every string ending with 01. Give the formal 5-tuple $(Q, \Sigma, \delta, q_0, F)$, the transition table and the state diagram, and trace the extended transition function (δ-hat) on the input 11001. (b) Design a DFA over $\Sigma = \{0, 1\}$ that accepts every string in which all binary strings are divisible by 3. Write the language of your DFA in set-builder notation.	3 (1.5+1.5)	CO1